

SPRY1 and the GRB2 SH3-Binding PxxPxR Motif: Hypothesis Evaluation

Target gene: SPRY1 (human, UniProt O43609) · **Term under evaluation:** protein sequestering activity (GO:0140311) · **Hypothesis slug:** `spry1-grb2-sh3-motif` · **Reference context:**

P 12402043

Summary

The seed hypothesis proposes that human SPRY1 (O43609) "carries a C-terminal PxxPxR GRB2 SH3-binding motif equivalent to the one that lets SPRY2 sequester GRB2," and asks for a single decisive analysis: scan human SPRY1, SPRY2, and SPRY4 plus their mouse and zebrafish orthologs for PxxPxR motifs and report positions and conservation. I ran that scan on the canonical UniProt sequences using the regex `P..P.R`. The result **refutes** the hypothesis. The canonical PxxPxR GRB2-SH3 ligand — **PTVPPR at position 304** — is present and conserved specifically in mammalian **SPRY2** (human O43597 and mouse Q9QXV8). It is **absent from SPRY1** (human O43609, mouse Q9QXV9), **absent from SPRY4** (human Q9C004, mouse Q9WTP2), and **absent from the C-terminus of zebrafish spry2** (Q4VBS7). Human SPRY1's actual C-terminal sequence (`...LESCPSRGQGKPS`) contains no PxxPxR motif at all.

This computational result is in exact agreement with the primary experimental literature, which describes the C-terminal PXXPX_R GRB2-SH3 motif as "**found exclusively on Spry2**" ([PMID:16893902](#)) and as an "**exclusive, necessary, but cryptic PXXPX_R motif in the C terminus of Spry2**" ([PMID:17255109](#)). Because the structural premise of the hypothesis is false, the proposed basis for assigning **GO:0140311 (protein sequestering activity)** to SPRY1 — GRB2 sequestration via a SPRY2-copied SH3-ligand motif — does not hold. SPRY1 should **not** inherit this annotation by paralog analogy to SPRY2.

Two caveats bound the verdict. First, "refuted" applies narrowly to the *specific mechanism named in the hypothesis* (a constitutive C-terminal SH3 ligand). SPRY1 does conserve an alternative GRB2-engagement feature: the N-terminal phospho-tyrosine Y53 in the context `NEYTEG`, which the supplied reference [PMID:12402043](#) identifies as the growth-factor-inducible tyrosine mediating SPRY1/SPRY2 binding to GRB2. Second, even for SPRY2 — the paralog that

does possess the motif — whether GRB2 binding is functionally required for ERK inhibition is actively contested ([PMID:17689925](#)). Both caveats *strengthen* rather than weaken the recommendation against a sequestration annotation for SPRY1.

Key Findings

Finding 1 — SPRY1 lacks the C-terminal PxxPxR GRB2 SH3-binding motif present in SPRY2

The hypothesis-specific test was a regex scan (`P..P.R`) of the canonical UniProt sequences for the three human paralogs and their mouse and zebrafish orthologs. The outcome is unambiguous:

Protein	UniProt	PxxPxR hits	C-terminal SPRY2-type motif?
Human SPRY2	O43597	PTVPPR @ pos 304	Yes (C-terminal)
Mouse Spry2	Q9QXV8	PTVPPR @ pos 304	Yes (conserved)
Human SPRY1	O43609	none	No
Mouse Spry1	Q9QXV9	none	No
Human SPRY4	Q9C004	none	No
Mouse Spry4	Q9WTP2	none	No
Zebrafish spry2	Q4VBS7	none at C-term (<code>...CRCKNKGVEK</code>)	No
Zebrafish spry1	B1PBZ7	N-terminal hits only (pos 34, 44)	No (not SPRY2-type)

Human SPRY1's C-terminal segment (`...LESCPSRGQGKPS`) contains no PxxPxR. The only canonical C-terminal PxxPxR GRB2-SH3 ligand in any mammalian Sprouty is SPRY2's PTVPPR at position 304, and it is conserved between human and mouse — exactly the paralog- and position-specific signature expected of a SPRY2-specific innovation. The scattered N-terminal "hits" in zebrafish spry1 (positions 34 and 44) are neither positionally nor functionally equivalent to the SPRY2 C-terminal SH3 ligand and do not rescue the hypothesis.

This matches the experimental record directly. [PMID:16893902](#) states that GRB2 binding "correlates with the binding to Grb2 via a C-terminal proline-rich sequence that is **found exclusively on Spry2**. This PXXPXR motif binds directly to the N-terminal Src homology domain 3 of Grb2, and when added onto the C terminus of Spry4 the resultant chimera inhibits the Ras/ERK pathway." The chimera experiment is especially telling: SPRY4 acquires the GRB2-SH3 activity *only when the SPRY2 motif is grafted onto it*, confirming that non-SPRY2 paralogs — SPRY4, and by the same logic SPRY1 — do not natively possess the motif. [PMID:17255109](#) independently reinforces exclusivity: "An exclusive, necessary, but cryptic PXXPXR motif in the C terminus of Spry2 is revealed upon stimulation." The core structural claim of the hypothesis is therefore false.

Finding 2 — Any SPRY1–GRB2 interaction uses a distinct phospho-tyrosine mechanism, and the functional relevance of GRB2 binding is contested even for SPRY2

The reference supplied with the hypothesis, [PMID:12402043](#), describes a mechanism *different* from the one the hypothesis invokes: "after stimulation by growth factors Spry1 and Spry2 translocate to the plasma membrane and become phosphorylated on a conserved tyrosine. Next, they bind to the adaptor protein Grb2." This is an **inducible, phospho-tyrosine-dependent** engagement — not a constitutive C-terminal SH3 ligand. In other words, to whatever extent SPRY1 binds GRB2 at all, it would do so through a route that has nothing to do with the PxxPxR motif named in the hypothesis.

Moreover, the functional significance of GRB2 binding is directly challenged even for SPRY2. [PMID:17689925](#) maps the SPRY2–GRB2 interface to the GRB2 N-terminal SH3 domain and two proline-rich stretches (residues 59–64 and 303–307) and then tests a GRB2-binding-dead double mutant: "a double Sprouty2 mutant (hSpry2 P59AP304A), which is unable to bind Grb2, developed at a similar inhibition level of fibroblast growth factor receptor-ERK pathway than that which originated from Sprouty2 wt. These results are evidence that the Sprouty2 mechanism of ERK inhibition is independent of Grb2 binding." If GRB2 binding is dispensable for ERK inhibition in the paralog that *does* possess the motif, then a sequestration-based functional annotation is on weak ground even in the best case — and weaker still for SPRY1, which lacks the motif entirely.

Finding 3 — SPRY1 conserves the N-terminal NEYTEG phospho-tyrosine (Y53) but none of SPRY2's canonical SH3-ligand proline motifs

Sequence inspection localizes the conserved regulatory tyrosine. In SPRY2 the key tyrosine sits in the motif `...RNTNEYTEGP...`; human SPRY1 conserves the identical `NEYTEG` context at **Tyr53** (`...RGSNEYTEGP...`), whereas SPRY4 has a divergent context (`...VENDYIDNP...`, Tyr52). This conserved N-terminal tyrosine is precisely the residue that

P 12402043

requires for growth-factor-induced GRB2 binding, and its presence in SPRY1 defines the *alternative*, non-SH3 route.

Importantly, the SPRY2 N-terminal GRB2-contact proline stretch (residues 59–64, `PTVVPR`) is itself **not** a canonical PxxPxR (`P..P.R`) motif, and SPRY1's aligned region (`PSVVKR`) is likewise not PxxPxR. Thus neither the N-terminal nor the C-terminal SPRY2 proline features are reproduced as a canonical SH3 ligand in SPRY1. The only canonical PxxPxR in any mammalian Sprouty remains SPRY2 position 304 (PTVPPR). The distinction is mechanistically essential: a conserved regulatory tyrosine is not evidence for a constitutive GRB2-SH3 sequestration activity.

Mechanistic Model / Interpretation

The two paralogs engage GRB2 — if at all — through architecturally different features:

	N-terminus		C-terminus
SPRY2 (043597):	<code>...RNT[NEYTEG]P...</code>	<code>...(PTVVPR 59-64)...</code>	<code>...[PTVPPR @304]...</code>
	[^] Y55 conserved phospho-Tyr	[^] prolines (not PxxPxR)	[^] CANONICAL PxxPxR GRB2 N-SH3 ligand (SPRY2-EXCLUSIVE)
SPRY1 (043609):	<code>...RGS[NEYTEG]P...</code>	<code>...(PSVVKR aligned)...</code>	<code>...LESCPSRGQGKPS</code>
	[^] Y53 conserved phospho-Tyr	[^] not PxxPxR	[^] NO PxxPxR MOTIF
SPRY4 (Q9C004):	<code>...VEN[DYIDNP]...</code>	<code>...</code>	<code>...NO PxxPxR MOTIF</code>
	[^] Y52 divergent context		

The hypothesis conflates two separable modules:

1. **The C-terminal PxxPxR → GRB2-SH3 "sequestration" module.** This is a SPRY2-specific innovation (PTVPPR @304), conserved to mouse but absent from SPRY1, SPRY4, and zebrafish spry2. It is the structural basis for the GO:0140311-style "sequester GRB2" narrative. **SPRY1 does not have it.**
2. **The N-terminal conserved phospho-tyrosine (NEYTEG) → inducible GRB2 engagement.** This is conserved in SPRY1 (Y53) and is the route described in the reference

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But it is inducible and phospho-dependent, not a constitutive SH3-ligand sequestration mechanism, and its functional output (ERK inhibition) has been shown to be GRB2-independent even in SPRY2.

The correct reading: the specific molecular feature the hypothesis asserts (a SPRY2-equivalent C-terminal SH3 ligand) **does not exist in SPRY1**. Whatever GRB2-related biology SPRY1 participates in operates through a different, tyrosine-phosphorylation-gated mechanism whose status as "protein sequestering activity" is unsupported by the evidence reviewed here.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence / limitations
This analysis (UniProt scan, P . . P . R)	Computational / evolutionary	Refutes	Does SPRY1 carry a C-terminal PxxPxR?	PxxPxR (PTVPPR @304) present only in SPRY2 (human + mouse); none in SPRY1, SPRY4, zebrafish spry2 C-term	Canonical UniProt O43609/O43597/Q9C004 + mouse/zebrafish orthologs	High for human/depends on canonical isoform regex definition
PMID:16893902	Direct assay + mutagenesis	Refutes (motif = SPRY2-exclusive)	Is the C-terminal GRB2-SH3 motif paralog-specific?	PXXPXK "found exclusively on Spry2"; grafting onto SPRY4 confers ERK inhibition	Mammalian cells, FGFR/ERK	High; in overexp
PMID:17255109	Direct assay	Refutes (SPRY2-exclusivity)	Is the C-terminal PXXPXK SPRY2-exclusive and conditional?	"exclusive, necessary, but cryptic PXXPXK motif in the C terminus of Spry2 is revealed upon stimulation"	FGFR-ERK, PP2A context	High; co-SPRY2 specific
PMID:12402043 (reference)	Interaction / mechanism	Qualifies	How do SPRY1/2 bind GRB2?	GRB2 binding via conserved phospho-tyrosine after membrane translocation — not a constitutive SH3 ligand	Ras/MAPK, growth-factor stimulation	High; de-alternate mechanism conserv SPRY1
PMID:17689925	Mutant phenotype	Competing	Is GRB2 binding required for	GRB2-binding-dead SPRY2 (P59A/P304A) still inhibits	FGFR-ERK, SPRY2 mutants	High; undercu sequest

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitation
			ERK inhibition?	FGFR-ERK → inhibition is GRB2-independent		as-function even for
PMID:24469046	Review/database + assay	Qualifies	Regulation of SPRY2-GRB2 binding	CK1 phosphorylation modulates SPRY2-GRB2 binding and ERK inhibition	FGF-ERK, PC12 / gastric cancer	Medium SPRY2-f orientation only
PMID:39456824	Review	Orientation	SPRY2 as GRB2-SOS inhibitor	SPRY2 binds GRB2, inhibits GRB2-SOS, attenuates RAS/ERK	Review	Low-medium review-SPRY2-c

Evidence Base

- **PMID:16893902** — *A Src homology 3-binding sequence on the C terminus of Sprouty2 is necessary for inhibition of the Ras/ERK pathway downstream of fibroblast growth factor receptor stimulation.* The keystone paper: it demonstrates the C-terminal PXXPXR GRB2-SH3 ligand is **exclusive to SPRY2**, binds GRB2's N-terminal SH3, and that transplanting it onto SPRY4 is sufficient to confer ERK inhibition. This directly establishes that SPRY1/SPRY4 lack the motif natively — exactly what the sequence scan found.
- **PMID:17255109** — *Direct binding of PP2A to Sprouty2...* Confirms the C-terminal PXXPXR motif is "exclusive" to SPRY2 and only becomes functionally "revealed" upon stimulation, reinforcing both exclusivity and conditionality.
- **PMID:12402043** — *Sprouty1 and Sprouty2 provide a control mechanism for the Ras/MAPK signalling pathway.* The hypothesis's own reference. It describes SPRY1/SPRY2 binding GRB2 via a conserved phospho-tyrosine after membrane translocation — a mechanism distinct from a C-terminal SH3 ligand and the basis for the alternative (non-refuted) route conserved at SPRY1 Y53.

- [PMID:17689925](#) — *Sprouty2 binds Grb2 at two different proline-rich regions, and the mechanism of ERK inhibition is independent of this interaction.* Competing evidence: a GRB2-binding-dead SPRY2 mutant still inhibits ERK, so GRB2 sequestration is not the functional mechanism even in SPRY2.
- [PMID:24469046](#) and [PMID:39456824](#) — Orientation/review-level material on SPRY2–GRB2 regulation and the GRB2-SOS inhibition model; both are SPRY2-centric and do not provide SPRY1-specific support for GO:0140311.

GO Curation Implications

Leads for curator verification — do not auto-apply:

- **GO:0140311 (protein sequestering activity) should NOT be assigned to SPRY1** on the basis of the SPRY2 PxxPxR/GRB2-SH3 paradigm. The structural premise (a SPRY2-equivalent C-terminal SH3 ligand in SPRY1) is directly refuted by the sequence scan and by primary literature stating the motif is SPRY2-exclusive. Assigning GO:0140311 to SPRY1 by analogy would be **paralog over-annotation** / database carry-over.
- If any GRB2-related SPRY1 annotation is contemplated, it must rest on **SPRY1-specific experimental evidence** for the phospho-tyrosine (Y53/NEYTEG) route (per

[P 12402043](#)

not on the C-terminal motif. Even then, the appropriate molecular-function term is uncertain because GRB2 binding is dispensable for ERK inhibition in SPRY2

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- **MF vs BP vs CC:** The best-supported SPRY1 role in this evidence set is participation in **negative regulation of RTK/RAS/MAPK signaling** (a BP-level concept), not a specific sequestration MF. For MF, avoid "protein sequestering activity" here; "protein binding" would be uninformative and should be avoided as a final recommendation. If an MF is required, tie it explicitly to the phospho-tyrosine mechanism and flag it as *SPRY1-specific-evidence-pending*.
- **Net recommendation:** Treat GO:0140311 as **non-core / unsupported for SPRY1**; retain it (where experimentally grounded) for **SPRY2**, not SPRY1.

Mechanistic Scope

The immediate molecular event under test is: *does SPRY1 directly present a C-terminal PxxPxR peptide that binds the GRB2 N-terminal SH3 domain to sequester GRB2?* The answer is no — the peptide does not exist in SPRY1. This is a statement about a **direct gene-product structural feature**, the cleanest level at which to evaluate the hypothesis.

Separable and downstream are: (i) SPRY1's broader role as a negative regulator of RAS/MAPK signaling (a pathway-level consequence), (ii) developmental phenotypes from *Spry1* loss of function, and (iii) any disease associations. None of these require the C-terminal PxxPxR motif, and none should be used to back-infer a GRB2-sequestration MF for SPRY1. The conserved N-terminal phospho-tyrosine (Y53) is a *direct* feature of SPRY1, but its engagement of GRB2 is inducible and its necessity for ERK inhibition is contested — so it supports, at most, a conditional interaction, not a constitutive sequestration activity.

Conflicts and Alternatives

- **Paralog confusion (primary risk):** The entire GRB2-SH3 "sequestration" story in the literature is SPRY2-specific. Transferring it to SPRY1 is exactly the over-annotation error the evidence warns against. The chimera experiment

(P 16893902),

where SPRY4 gains the activity only after the SPRY2 motif is grafted on, is a direct demonstration that non-SPRY2 paralogs lack the motif natively. (During accession lookup, even the mouse *Spry1*/*Spry2* entries were easy to swap, underscoring this risk.)

- **Mechanism substitution:** The reference

P 12402043

describes a phospho-tyrosine route, not an SH3-ligand route. A curator could misread "SPRY1 binds GRB2" (true, inducibly) as "SPRY1 sequesters GRB2 via PxxPxR" (false).

- **Function vs. binding:**

P 17689925

shows GRB2 binding is not required for SPRY2's ERK inhibition, so even a confirmed SPRY1–GRB2 interaction would not establish sequestration as SPRY1's functional mechanism.

- **Organism specificity:** Zebrafish spry2 (Q4VBS7) lacks the C-terminal PxxPxR; zebrafish spry1 (B1PBZ7) has only N-terminal PxxPxR-like stretches. The motif is thus neither pan-vertebrate nor SPRY1-specific.
- **Isoform caveat:** The scan used canonical UniProt sequences; non-canonical SPRY1 isoforms were not exhaustively examined.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	How to resolve
Isoform coverage	Canonical UniProt O43609 only	A rare SPRY1 isoform could in principle contain a C-terminal proline insertion	Scan all annotated SPRY1 isoforms + Ensembl transcripts
Direct SPRY1–GRB2 assay	Relied on joint SPRY1/SPRY2 statement in P 12402043	Strength of SPRY1-specific (not SPRY2) GRB2 binding is not quantified here	SPRY1-specific co-IP / ITC with GRB2 ± Y53 mutation
Non-canonical SH3 ligands	Only canonical PxxPxR (P . . P . R) regex used	GRB2 SH3 can tolerate atypical ligands; a non-PxxPxR SPRY1 ligand could be missed	SPOT/peptide-array or structural docking against GRB2 N-SH3
Functional output	Took P 17689925 (SPRY2) as read-across	SPRY1 ERK-inhibition dependence on GRB2 not directly tested	SPRY1 GRB2-binding-dead mutant ERK rescue assay
GO:0140311 applicability	Evaluated against the PxxPxR premise only	The term might in principle apply via a different SPRY1 partner	Curate against all SPRY1 interaction evidence, not just GRB2

Discriminating Tests

1. **Exhaustive isoform/transcript scan** of SPRY1 (all UniProt isoforms + Ensembl) for any C-terminal proline-rich/PxxPxR feature — fastest way to fully close the structural question.
 2. **SPRY1-specific GRB2 co-immunoprecipitation** with and without Y53 (NEYTEG) mutated, under growth-factor stimulation, to confirm the phospho-tyrosine route and exclude an SH3 route.
 3. **GRB2 N-SH3 pulldown / peptide-array** using SPRY1 C-terminal peptides vs. the SPRY2 PTVPPR peptide as positive control — a direct binding discriminator.
 4. **SPRY1 GRB2-binding-dead mutant ERK rescue**, mirroring the SPRY2 P59A/P304A experiment, to test whether any SPRY1 function depends on GRB2.
 5. **Cross-species ortholog alignment** anchored on SPRY2 position 304 to formally confirm the motif's SPRY2-restricted phylogenetic distribution.
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Proposed Follow-up Actions / Curation Leads

Leads requiring curator verification:

- **Action change:** Do not assign **GO:0140311 (protein sequestering activity)** to SPRY1 based on the PxxPxR/GRB2-SH3 analogy. Flag the premise as refuted.
- **Candidate reference + snippet:** [PMID:16893902](#) — *"via a C-terminal proline-rich sequence that is found exclusively on Spry2. This PXXPXR motif binds directly to the N-terminal Src homology domain 3 of Grb2."*
- **Candidate reference + snippet:** [PMID:17255109](#) — *"An exclusive, necessary, but cryptic PXXPXR motif in the C terminus of Spry2 is revealed upon stimulation."*
- **Candidate reference + snippet (alternative mechanism):** [PMID:12402043](#) — *"Spry1 and Spry2 translocate to the plasma membrane and become phosphorylated on a conserved tyrosine. Next, they bind to the adaptor protein Grb2."*
- **Candidate reference + snippet (functional caveat):** [PMID:17689925](#) — *"a double Sprouty2 mutant (hSpry2 P59AP304A), which is unable to bind Grb2... the Sprouty2 mechanism of ERK inhibition is independent of Grb2 binding."*
- **Suggested curator question:** Is the SPRY1 GRB2/sequestration lineage derived from SPRY2 by paralog inference? If so, flag for removal or qualification.

- **Suggested experiments:** SPRY1 Y53 mutant GRB2 co-IP; SPRY1 C-terminal peptide vs. GRB2 N-SH3 pulldown (tests 2–3 above).
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Conclusion

The hypothesis that human SPRY1 carries a C-terminal PxxPxR GRB2 SH3-binding motif "equivalent to the one that lets SPRY2 sequester GRB2" is **refuted**. The canonical PxxPxR motif (PTVPPR @304) is a SPRY2-specific, mouse-conserved feature absent from SPRY1, SPRY4, and the C-terminus of zebrafish spry2, in exact agreement with primary literature describing the motif as "found exclusively on Spry2." SPRY1 instead conserves an N-terminal phospho-tyrosine (Y53/NEYTEG) that mediates an inducible, mechanistically distinct GRB2 engagement — a route whose functional necessity is contested even for SPRY2. Accordingly, **GO:0140311 (protein sequestering activity) should not be assigned to SPRY1 on the basis of this hypothesis**, and any GRB2-related SPRY1 annotation must await SPRY1-specific experimental evidence for the phospho-tyrosine mechanism.

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